

🧠 projectPal Lab #001: Agile vs Waterfall Deep Dive

Lab Type: Idea

Estimated Time: 30–45 mins

Skill Level: Beginner

```
# Let's begin by printing your name to personalize the notebook
your_name = "Beryl Eduah"
print(f"Welcome to the lab, {your_name}!")
```

Welcome to the lab, Beryl Eduah!

🔍 STAR Method Lab Prompt

Waterfall Model (Linear and sequential)

How it works

Requirements → Design → Build → Test → Deploy → Maintain

Each phase must be completed before moving to the next.

Agile Model (Iterative and incremental)

How it works

Project is broken into small cycles called iterations or sprints.

Continuous planning, development, testing, and feedback.

Situation:

A fintech startup needs to build a mobile banking app. The market is volatile, and user requirements are expected to change frequently based on feedback.

Task:

Determine whether to use a linear (Waterfall) or iterative (Agile) approach to ensure the product stays relevant.

Action:

Choose Agile (Scrum). Set up 2-week "Sprints." Instead of building the whole app at once, focus on a "Minimum Viable Product" (MVP) that only allows users to check their balance.

Expected Result:

The team releases the MVP in 6 weeks. User feedback shows they prioritize "Quick Transfers" over "Profile Customization," allowing the team to pivot the roadmap immediately.

🔧 Your Assignment

Use this section to complete your deliverable:

- **Methods Compared:** Agile vs Waterfall
- **Use Case:** Mobile Banking App for Finetech Startup
- **Gemini Prompt:** Pros and cons table + STAR justification

Waterfall Method:

- Risk of releasing an outdated product after 6–12 months.
- Failures are often discovered late in the testing phase.
- Fixed cost based on set requirements; changes are expensive.
- Value is delivered only at the very end of the project.

Agile Method:

- High; features are adjusted based on current market trends.
- Issues are identified early during bi-weekly sprint reviews.
- Flexible; budget is spent on the highest-priority features.
- Value is delivered early through an MVP (Minimum Viable Product).
- **STAR Output:**
 - S: Fintech startup needs a banking app but fears building obsolete features in a volatile market.
 - T: Choose a methodology to ensure product relevance and a steady release cadence.

- A: Used Gemini to analyze models; selected Agile/Scrum with 2-week sprints focused on a balance-checking MVP.
- R: MVP launched in 6 weeks. Real-time data prioritized "Quick Transfers" over "Profiles," saving costs by pivoting the roadmap early.